



RTR (RADIO TELEPHONY)

DGCA
UPDATE 2.0

QUESTION BANK



1. What is the internationally recognized Distress signal?

- (A) PAN-PAN (spoken three times)
- (B) SECURITE (spoken three times)
- (C) MAYDAY (spoken three times)
- (D) EMERGENCY (spoken once)

MAYDAY
MAYDAY
MAYDAY



2. What is the internationally recognized Urgency signal?

- (A) PAN-PAN (spoken three times)
- (B) MAYDAY (spoken three times)
- (C) EMERGENCY (spoken three times)
- (D) SECURITE (spoken once)

PAN-PAN
PAN-PAN
PAN-PAN



3. The meaning of the abbreviation QNH is:

- (A) Altimeter setting to read height above the aerodrome.
- (B) Altimeter setting to read height above Mean Sea Level (MSL).
- (C) The standard pressure of 1013.2 hPa.
- (D) Pressure at the station elevation.



4. The meaning of the abbreviation QFE is:

- (A) Altimeter setting to read height above MSL.
- (B) Standard atmospheric pressure (1013.2 hPa).
- (C) Altimeter setting to read aerodrome elevation when on the ground.
- (D) Expected time en route.



5. What altimeter setting must be used when operating at or above the Transition Level (TL)?

- (A) QFE
- (B) QNH of the destination aerodrome
- (C) The standard pressure setting (1013.2 hPa)
- (D) The lowest QNH along the route



6. Which of the following is NOT an essential piece of information for a standard Position Report?

- (A) Time over the reporting point.
- (B) Position/Identification of the reporting point.
- (C) Fuel on board in gallons.
- (D) Next significant point and time estimate.



7. What action must an aircraft take immediately upon hearing the signal "MAYDAY" on a frequency it is monitoring?

- (A) Reply to the distressed aircraft to offer assistance.
- (B) Cease all transmissions immediately and maintain a listening watch.
- (C) Transmit "SEELONCE FEENEE" to clear the frequency.
- (D) Change frequency to avoid congestion.

MAYDAY
MAYDAY
MAYDAY



8. Define the term Readability Code 3.

- (A) Unreadable
- (B) Readable now and then
- (C) Readable, but with difficulty
- (D) Perfectly readable



9. How must the number "12,500" be spoken in radiotelephony?

- (A) ONE TWO FIVE ZERO ZERO
- (B) ONE TWO THOUSAND FIVE HUNDRED
- (C) TWELVE THOUSAND FIVE HUNDRED
- (D) ONE TWO DECIMAL FIVE ZERO ZERO



10. Which instruction from ATC must be read back by the pilot?

- (A) Take-off clearance
- (B) ATC route clearances
- (C) Advisory information
- (D) Wind speed and direction



SUMMARY

- ★ Use correct distress and urgency signals.
- ★ Know altimeter settings: QNH, QFE, QNH.
- ★ Follow correct phraseology and read back ATC instructions.
- ★ Maintain clear, concise and accurate radiotelephony communication for safe operations.



REMEMBER

- ★ MAYDAY - Distress
- ★ PAN-PAN - Urgency
- ★ SECURITE - Safety
- ★ Read back important ATC instructions.
- ★ Listen, acknowledge and assist when required.





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11. What is the full phraseology for cancelling an urgent message if the situation has improved?

- (A) PAN-PAN TERMINATED
- (B) CANCELLING URGENCY
- (C) PAN-PAN OVER
- (D) SECURITE ENDED

PAN-PAN
TERMINATED



12. Under what condition is an aircraft permitted to use a shortened callsign?

- (A) The aircraft initiates the shortening after the first successful contact.
- (B) When the pilot deems the radio channel to be non-congested.
- (C) After the ground station (ATC) has initiated the use of the shortened callsign.
- (D) Never, as ICAO prohibits abbreviated callsigns.



13. What is the standard VHF Emergency Frequency?

- (A) 121.5 MHz
- (B) 123.45 MHz
- (C) 118.0 MHz
- (D) 122.8 MHz



14. Which are the two highest priority messages after Distress and Urgency calls?

- (A) Movement and Safety.
- (B) Meteorological and Regularity.
- (C) Safety and Regularity.
- (D) Positioning and Administrative.



15. If a message is classified as a Safety message, what does it primarily concern?

- (A) The safety of the destination runway.
- (B) The safety of navigation or the significant weather warnings.
- (C) The regular operational movements of the aircraft.
- (D) Administrative communication between ground stations.



16. What is the meaning of the phrase "SQUAWK 7700"?

- (A) Radio Communication Failure
- (B) Unlawful Interference
- (C) General Emergency
- (D) VFR flight below 10,000 feet



17. What is the meaning of the phrase "SQUAWK 7600"?

- (A) General Emergency
- (B) Unlawful Interference
- (C) Radio Communication Failure
- (D) Low fuel status



18. What is the correct phraseology for requesting permission to change frequency?

- (A) REQUEST CHANGE TO [frequency/station].
- (B) CLEARED TO CHANGE FREQUENCY.
- (C) ROGER CHANGE FREQUENCY.
- (D) Switching to [frequency/station] now.



19. What is the primary purpose of Selective Calling (SELCAL)?

- (A) To automatically select the best radio channel.
- (B) To provide an alternate communication means during RCF.
- (C) To alert a specific aircraft crew to an incoming radio call.
- (D) To encrypt radio communication for security.



20. What is the full form of the abbreviation METAR and what information does it contain?

- (B) Medium Terminal Advisory Report; forecast weather.
- (C) Aviation Routine Weather Report; actual weather observation.
- (C) Meteorological Terminal Area Report; general climate info.
- (D) Maximum Estimated Time and Altitude Required; flight path data.



21. What must an aircraft do if it experiences a complete Radio Communication Failure (RCF) under IFR in VMC?

- (A) Continue under IFR and try all available frequencies.

- (B) Squawk 7600 and follow the lost communications procedures.
- (C) Squawk 7700 and continue the flight plan as filed.
- (D) Land at the nearest aerodrome regardless of conditions.



REMEMBER

- ★ Distress > Urgency > Safety > Regularity.
- ★ Use correct phraseology and brevity.
- ★ Listen, acknowledge and read back.
- ★ Clear communication ensures safety.



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21. What must an aircraft do if it experiences a complete Radio Communication Failure (RCF) under IFR in VMC?

- (A) Climb to the highest minimum safe altitude.
- (B) Continue the flight in VMC, land at the nearest suitable aerodrome, and report by phone.
- (C) Proceed according to the last received clearance under IFR rules.
- (D) Squawk 7700 and hold over the nearest VOR.



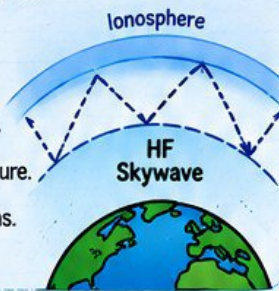
22. If experiencing RCF in IMC, the pilot should immediately set the transponder to which code?

- (A) 7700
- (B) 7600
- (C) 7500
- (D) 2000



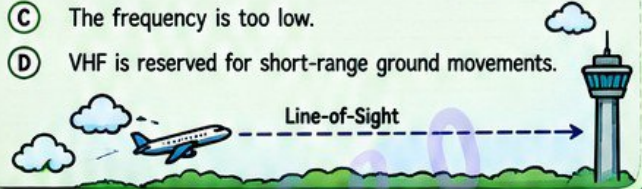
23. Which radio wave property allows HF to travel long distances?

- (A) Line-of-sight propagation.
- (B) Skywave reflection of the Ionosphere.
- (C) Ground wave following the Earth's curvature.
- (D) Ducting caused by temperature inversions.



24. Why is VHF communication primarily line-of-sight?

- (A) The transmitter power is very low.
- (B) VHF waves do not bend significantly around the Earth's curvature or reflect off the ionosphere.
- (C) The frequency is too low.
- (D) VHF is reserved for short-range ground movements.



25. What does the term 'Simplex' operation mean in radiotelephony?

- (A) Two frequencies used simultaneously for both transmission and reception.
- (B) A single frequency used for both transmission and reception, but only one station transmits at a time.
- (C) The ability to transmit and receive simultaneously on a single frequency.
- (D) Communication that requires a repeater station.



26. What type of modulation is used for standard VHF voice communication in aviation?

- (A) Amplitude Modulation (AM)
- (B) Frequency Modulation (FM).
- (C) Single Sideband (SSB)
- (D) Pulse Code Modulation (PCM)



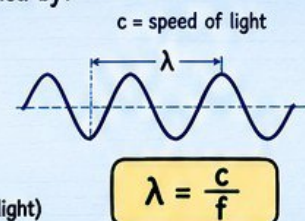
27. What is the function of the Squelch circuit in a radio receiver?

- (A) To increase the volume of weak signals.
- (B) To tune the receiver to the exact frequency.
- (C) To suppress background receiver noise when no signal is present.
- (D) To automatically select the best antenna.



28. The relationship between Wavelength (λ) and Frequency (f) is defined by?

- (A) $\lambda = f / c$
- (B) $\lambda = f \times c$
- (C) $\lambda = c - f$
- (D) $\lambda = c / f$ (where c is the speed of light)



29. What is the primary advantage of using the Single Sideband (SSB) technique in HF communication?

- (A) Superior clarity and audio fidelity.
- (B) Elimination of all static and noise.
- (C) Increased range and efficiency by reducing bandwidth and wasted power.
- (D) Easier setup and tuning compared to AM.



30. What is the function of the Transponder in air traffic control?

- (A) To transmit radio messages to ATC.
- (B) To reply to ground radar interrogation with a code and altitude information.
- (C) To receive all ATC instructions and display them digitally.
- (D) To automatically record all radio transmissions.



KEY TAKEAWAY

- ✓ HF = Ionospheric reflection → Long range
- ✓ VHF = Line-of-sight → Short/medium range
- ✓ SSB = More range & efficiency (HF)
- ✓ Simplex = One station at a time
- ✓ Squelch = Removes background noise

REMEMBER

- ★ 7700 = General Emergency
- ★ 7600 = Radio Communication Failure
- ★ 7500 = Unlawful Interference
- ★ VHF voice = FM
- ★ METAR = Actual weather observation





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31. When setting up a radio, what step must be taken before speaking into the microphone?

- (A) Key the microphone briefly before speaking.
- (B) Wait 5 seconds after keying to allow the transmitter to stabilize.
- (C) Check the volume level.
- (D) Ensure the squelch is turned off.



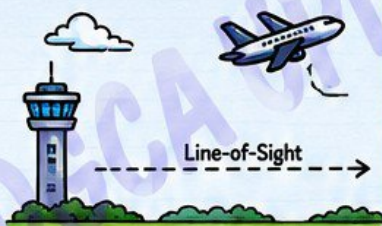
32. The process of separating the voice signal from the carrier wave in the receiver is called:

- (A) Modulation
- (B) **Demodulation**
- (C) Rectification
- (D) Amplification



33. The most common radio wave propagation mode for VHF is:

- (A) Skywave.
- (B) Surface wave.
- (C) **Direct/Line-of-sight.**
- (D) Ionospheric scatter.



34. The term 'Dead Zone' is a problem primarily associated with which frequency band?

- (A) **VHF**
- (B) UHF
- (C) HF
- (D) SHF



35. What is the recommended maximum duration for a single transmission on the air?

- (A) No more than 60 seconds.
- (B) No more than 30 seconds.
- (C) **Only enough time to say the critical information.**
- (D) The pilot can transmit as long as necessary.



36. The speed of a radio wave in a vacuum is approximately:

- (A) 300 meters per second
- (B) 300 kilometers per second
- (C) **300,000 kilometers per second**
- (D) 30,000 kilometers per hour

C = Speed of light
 $\approx 3 \times 10^8 \text{ m/s}$
 $\approx 300,000 \text{ km/s}$

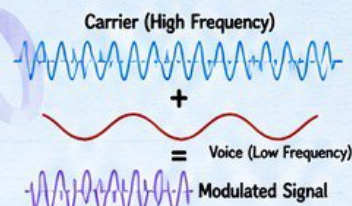
37. What is the primary limitation of VHF (Very High Frequency) radio for long-distance communication?

- (A) Susceptibility to sunspot activity.
- (B) Limited frequency bandwidth.
- (C) **Line-of-sight propagation.**
- (D) High power consumption.



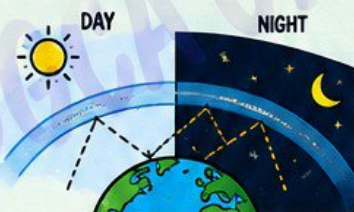
38. The process of modifying a high-frequency carrier wave with a low-frequency signal (voice) is called:

- (A) Demodulation
- (B) Coupling
- (C) **Modulation**
- (D) Rectification



39. Which band of frequencies is most affected by the changes in the Ionosphere between day and night?

- (A) VHF
- (B) UHF
- (C) **HF**
- (D) SHF



40. What is a Transceiver?

- (A) A device that only transmits radio signals.
- (B) A device that only receives radio signals.
- (C) **A unit combining a transmitter and a receiver.**
- (D) None of the above.



KEY TAKEAWAY



- ✓ Always follow proper radio procedures.
- ✓ Use only the necessary words.
- ✓ VHF = Line-of-sight → Limited range
- ✓ HF = Ionosphere dependent → Long range
- ✓ Clear communication = Safe operation

REMEMBER

- ★ Modulation → Transmit Demodulation → Receive
- ★ VHF → Line-of-sight (Up to Horizon)
- ★ HF → Affected by Ionosphere (Day & Night)
- ★ Radio wave speed (c) $\approx 3 \times 10^8 \text{ m/s}$
- ★ Transceiver → Transmit + Receive





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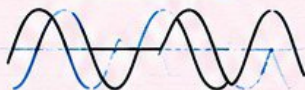


- 41.** In the formula for wavelength (λ) and frequency (f), $\lambda = c/f$, what does 'c' represent?

- (A) Capacitance
- (B) Speed of the aircraft
- (C) Speed of sound
- (D) Speed of light (radio wave)

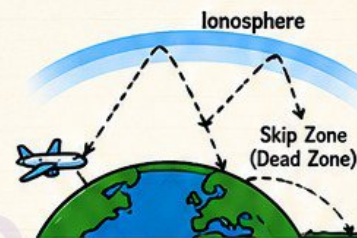
$$\lambda = \frac{c}{f}$$

c = speed of light
(radio wave)



- 42.** The term 'Dead Zone' or 'Skip Zone' is associated with which type of radio wave propagation?

- (A) Ground wave
- (B) Line-of-sight
- (C) Skywave (HF)
- (D) Tropospheric scatter



- 43.** Which type of antenna is commonly used for VHF communication on aircraft?

- (A) Long wire antenna
- (B) Blade or whip antenna
- (C) Loop antenna
- (D) Dish antenna



- 44.** What is the standard audio frequency range for human speech transmitted in radiotelephony?

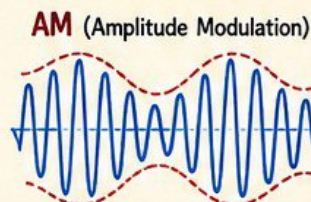
- (A) 100 kHz to 300 kHz
- (B) 300 Hz to 3400 Hz
- (C) 10 kHz to 20 kHz
- (D) 1 MHz to 5 MHz

Voice Frequency Range
300 Hz to 3400 Hz
(3.4 kHz)



- 45.** What does AM stand for in the context of radio modulation?

- (A) Antenna Modulation
- (B) Amplitude Modulation
- (C) Analog Modulation
- (D) Airborne Modulation



- 46.** The main advantage of using HF SSB (Single Sideband) for long-range communication over traditional AM is:

- (A) Lower bandwidth and higher range for the same power.
- (B) Higher fidelity and clearer audio.
- (C) It is immune to lightning and atmospheric static.
- (D) It only requires a small loop antenna.



- 47.** When two stations use two different frequencies for simultaneous two-way communication, this is known as:

- (A) Simplex
- (B) Half Duplex
- (C) Full Duplex
- (D) Triplex



- 48.** In an aircraft's VHF radio system, the inter-communication system used for communication between the cockpit and the cabin is called:

- (A) ATC Radio
- (B) Interphone or Cabin Announcing System
- (C) HF Radio
- (D) Transponder



- 49.** The electrical unit for measuring the rate of flow of current is the:

- (A) Volt
- (B) Ohm
- (C) Watt
- (D) Ampere



Ampere (A)
= Unit of Current

- 50.** What is the main purpose of an Automatic Voice Announcing System (AVAS) at an aerodrome?

- (A) To announce ATC instructions to pilots.
- (B) To inform pilots of wind conditions.
- (C) To provide automated voice information to pilots and passengers.
- (D) To transmit emergency alerts only.



KEY TAKEAWAY

- ✓ c in $\lambda = c/f$ is speed of light (radio wave).
- ✓ Dead Zone is related to Skywave (HF) propagation.
- ✓ VHF uses Blade/Whip antennas (line-of-sight).
- ✓ Voice frequency range: 300 Hz – 3400 Hz.
- ✓ AM = Amplitude Modulation.
- ✓ SSB is efficient: Lower bandwidth, higher range.
- ✓ Full Duplex uses two different frequencies.
- ✓ Interphone/CAS connects cockpit and cabin.
- ✓ Ampere (A) measures electric current.
- ✓ AVAS provides automated voice information.

REMEMBER

- ★ $\lambda = c/f \rightarrow c$ = speed of light (radio wave)
- ★ Dead Zone (Skip Zone) $\rightarrow$ Skywave (HF)
- ★ VHF $\rightarrow$ Line-of-sight communication
- ★ Voice range $\rightarrow$ 300 Hz to 3400 Hz (3.4 kHz)
- ★ AM $\rightarrow$ Amplitude Modulation
- ★ SSB $\rightarrow$ Saves power & bandwidth
- ★ Full Duplex $\rightarrow$ Two frequencies for two-way
- ★ AVAS $\rightarrow$ Automated Voice information

